

Atomistic Fréchet Distance (AFD)

Method, Sensitivity Analysis, and Results

Atomistic Evaluation

September 23, 2026

Abstract

AFD is a distributional, feature-based score for 3D atomistic generative models — the Fréchet Inception Distance recipe with the Inception network replaced by a pretrained chemistry-aware encoder. This report summarizes the method, a validation/sensitivity study establishing when AFD is statistically valid and how strongly it reacts to controlled degradations, and current AFD scores for a panel of published molecule and materials generators alongside preliminary legacy metrics.

1 AFD: Method

AFD (*Atomistic Fréchet Distance*) is “FID for atoms”: the same recipe as the Fréchet Inception Distance used to score image generators, with the Inception-v3 image encoder replaced by a **Conditional Equivariant Transformer (CT)** pretrained on atomistic data. Every structure — reference and generated — is embedded through the frozen CT, harvesting its 256-dimensional graph-level embedding (`mol_emb`). A Gaussian (μ, Σ) is fit to each set of embeddings, and AFD reports the closed-form Fréchet (Wasserstein-2) distance between the two Gaussians:

$$\text{AFD}^2 = \|\mu_r - \mu_g\|_2^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right), \quad (1)$$

where (μ_r, Σ_r) and (μ_g, Σ_g) are the reference and generated embedding statistics. **Lower is better**; $\text{AFD} = 0$ means the generated set is distributionally indistinguishable from the reference *in the CT feature space*. The point of this distribution-matching formulation is that AFD is a single, *set-level* score: it does not require per-sample labels, one-to-one matching, or a tractable likelihood. It asks only how faithfully the entire generated distribution reproduces a known ground-truth distribution in a chemistry-aware feature space, capturing drift in geometry, in composition, or in both at once.

The Fréchet form is attractive because it is closed-form and cheap — it needs only the first two moments of each feature set and reduces to a matrix square root — yet it still registers both a shift in the mean embedding and a change in the covariance (spread) of the distribution. Its shortcomings follow from the same assumption: modeling each feature set as a single Gaussian keeps only those first two moments, so AFD is blind to higher-order distributional structure, and the estimator is *biased upward* with a bias that grows as N shrinks — scores at different sample sizes are not comparable, and below the 256-d feature rank the covariance is degenerate and the number is meaningless (see Section 2). A second class of limitations comes from building the metric on **features from a pretrained model**: the checkpoint is part of the metric, so two AFD numbers are only comparable if produced with the same checkpoint against the same reference,

and the domain-matched checkpoint must be used (an in-domain encoder is dramatically more sensitive than a cross-domain one, Section 2.5). The score also inherits the encoder’s invariances — the CT embedding is E(3)-invariant and effectively intensive, so AFD is blind to chirality and to unit-cell multiplicity — and, being a single distributional number, it neither localizes which atoms drifted nor certifies that any individual structure is physically valid. AFD is therefore a comparative, distributional metric to be paired with validity/stability/novelty checks, not a replacement for them.

2 AFD Sensitivity Analysis

A sensitivity analysis establishes *when* AFD is a trustworthy measure and *how* it responds to known changes. It answers three questions: (i) how many samples are needed before a score is statistically valid and rankable — the finite- N validity floor; (ii) how small a difference is resolvable at a given N — the sensitivity floor; and (iii) how monotonically and how strongly the score reacts to controlled perturbations (position noise, atom mutation, vacancies, bond/lattice scaling), which both validates the metric and lets us calibrate a raw AFD value into physical units. Every experiment below compares a ground-truth reference against a controlled degradation of held-out ground-truth folds (fixed permutation, seed 2026), with the reference subsampled to match the target N on both sides; bands are 95% CIs over replicate folds and ρ denotes the Spearman rank correlation. Runtimes are wall-clock on one RTX A5000.

2.1 Sample size and the validity floor

Figures 1 and 2 measure pure finite- N bias: both sides are *unperturbed* ground truth, so the true distance is zero and any positive AFD is estimator bias. Panel (a) plots the AFD of each disjoint fold against its size N (log–log), with the per- N mean marked and the 256-d feature-dimension line; below that rank the covariance is singular and the small- N values ($N = 10, 100$) are artifacts. Panel (b) zooms into the full-rank regime ($N \geq 500$) and fits the floor as $\text{AFD} \approx \text{intercept} + c/N$, showing that once the covariance is well resolved the bias decays as $1/N$ toward zero. For QM9 the floor is 0.0105 at $N=500$, 0.0050 at $N=1,000$, 5.5×10^{-4} at $N=5,000$, and ≈ 0 at $N=10,000$; mp20 follows the same law (0.022, 0.011, 0.0015 at $N=500/1,000/5,000$). The practical rules: never compare AFD across different N ; use $N \geq 500$ for any meaningful score, $N \geq 1,000$ to rank models, and $N \geq 5,000$ to resolve gaps of ~ 0.001 between strong models.

AFD fold-vs-fold baseline & minimum sample size — qm9

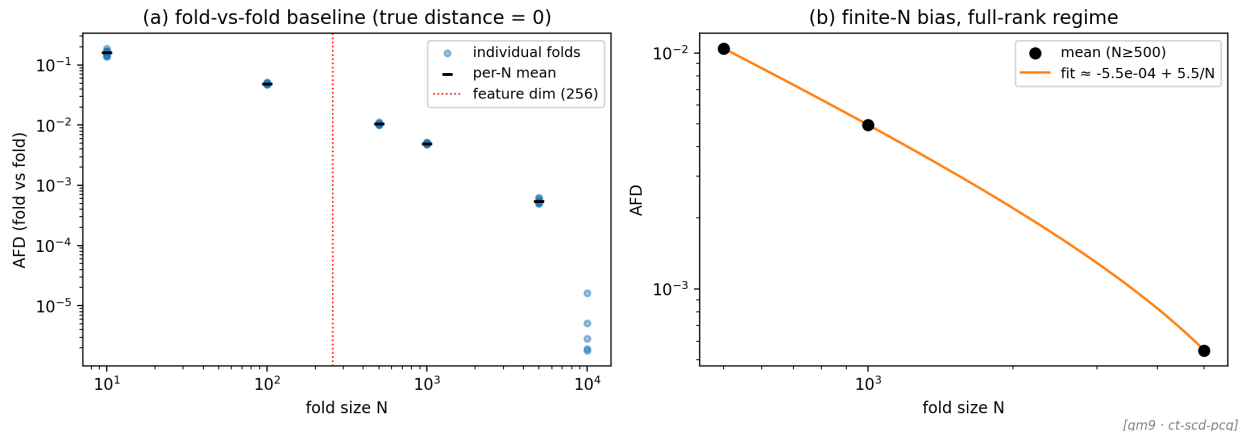


Figure 1: QM9 fold-vs-fold baseline (ct-scd-pcq). (a) AFD of individual disjoint ground-truth folds vs. fold size N with per- N means; the covariance is rank-degenerate left of the 256-d feature line. (b) The full-rank ($N \geq 500$) floor decays as $1/N$. Re-plotted from the raw `re_val` results without the mean connecting line, which the log axis could not draw where the $N=10,000$ mean reaches the numerical floor.

AFD fold-vs-fold baseline & minimum sample size — mp20

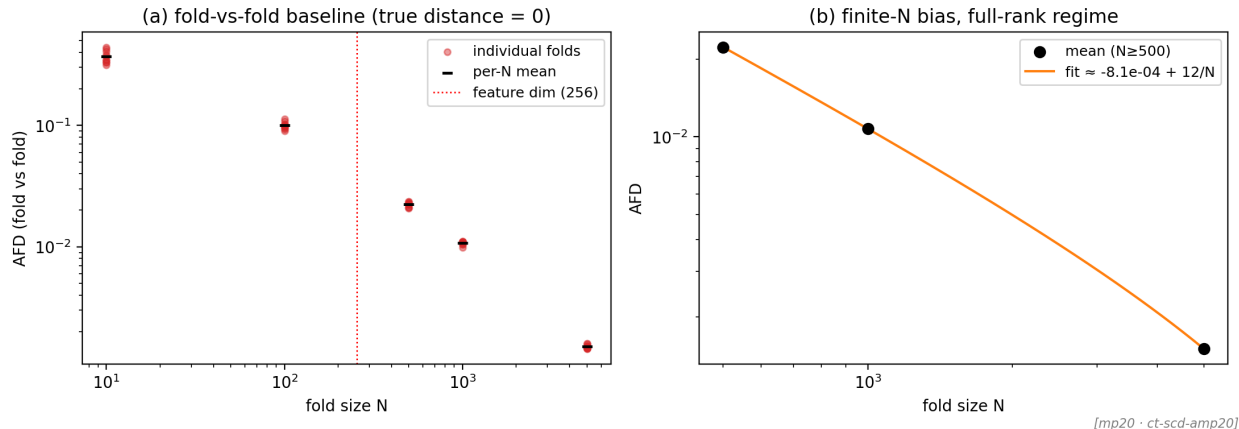


Figure 2: Materials (mp20, ct-scd-amp20) counterpart of Figure 1: the same rank floor and $1/N$ finite- N decay.

2.2 Molecular sensitivity (QM9)

The QM9 ladders (Figure 3) confirm AFD is monotone and sensitive to every degradation tested. Position noise sits at the floor for $\sigma \leq 0.01$ Å, is resolved by $\sigma \approx 0.03$ Å (0.149), and saturates near 0.44 by $\sigma \approx 0.1$ Å ($\rho = 0.95$); a *single* mutated atom scores 0.187 in-set / 0.289 for an out-of-distribution (OOD) element, and a single vacancy scores 0.196 ($\rho = 1.00$ for mutation and vacancy). Crucially, OOD chemistry (elements absent from the reference) scores higher than in-set changes at every count. The fine-grid noise sweep (Figure 4) and the alchemical experiment (Figure 5) reinforce this: the in-set mutation ladder rises monotonically (1 atom $\rightarrow$ 0.19, 5 $\rightarrow$ 0.38), and among directed single-atom swaps the OOD C $\rightarrow$ Si is the largest (≈ 0.37). Two useful negative controls appear in

Figure 6: bond-length scaling is strongly registered and V-shaped about 1.0 ($\rho = 0.97$), but a molecule and its **mirror image** score identically (0.0060 vs. 0.0061) — AFD is blind to chirality and must be paired with an explicit handedness check.

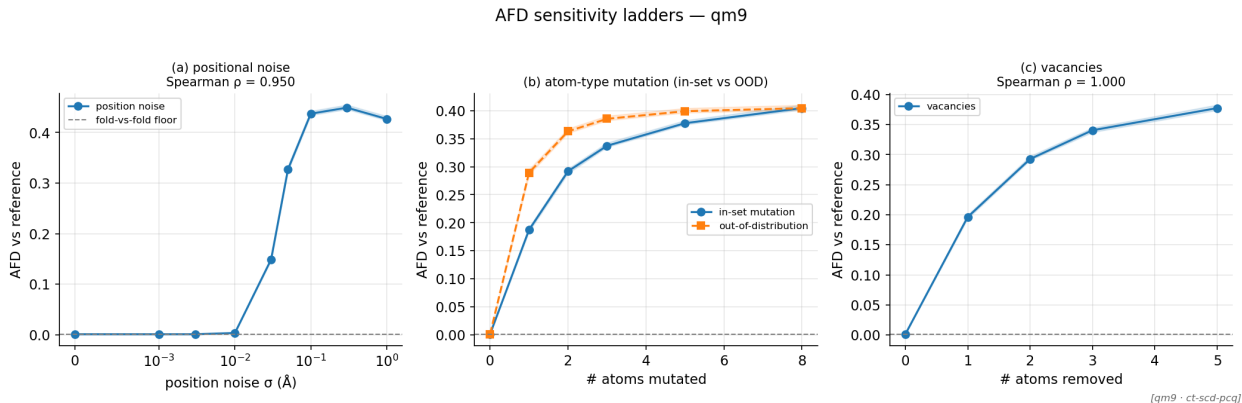


Figure 3: QM9 sensitivity ladders (mean + 95% CI vs. the fold floor): (a) position noise σ ; (b) atom-type mutation, in-set (solid) vs. OOD (dashed); (c) vacancies. All monotone; OOD > in-distribution at every count.

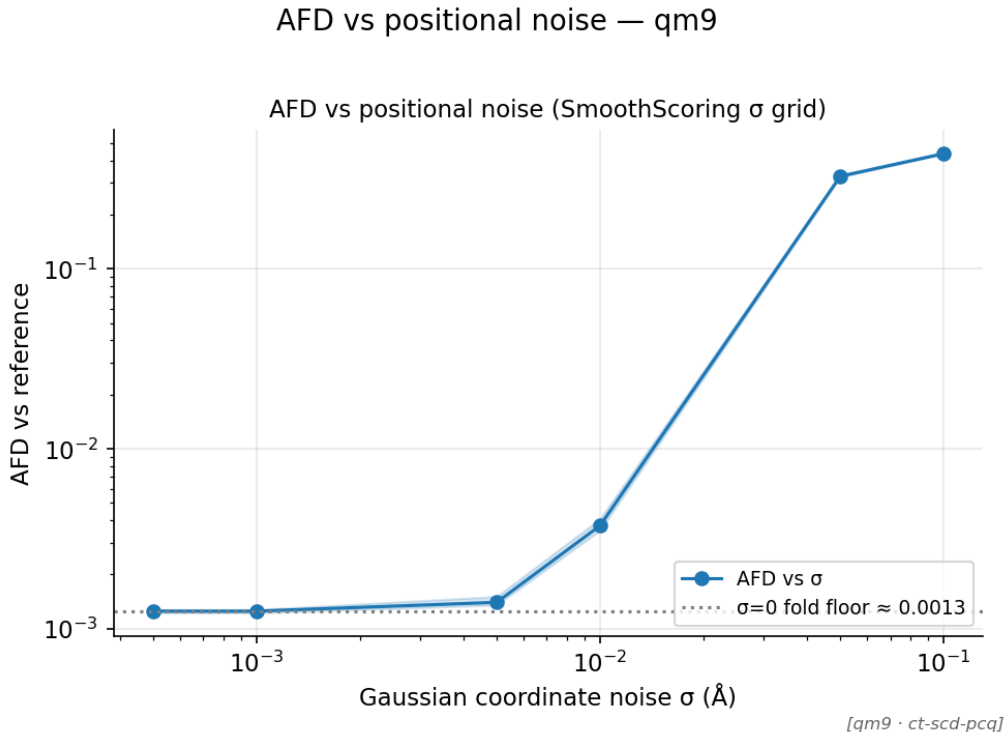


Figure 4: QM9 AFD vs. Gaussian coordinate noise on a fine σ grid (log-log): at the floor for $\sigma \leq 0.001 \text{ \AA}$, climbing from $\sigma \approx 0.005\text{--}0.01 \text{ \AA}$ to ≈ 0.44 at $\sigma = 0.1 \text{ \AA}$.

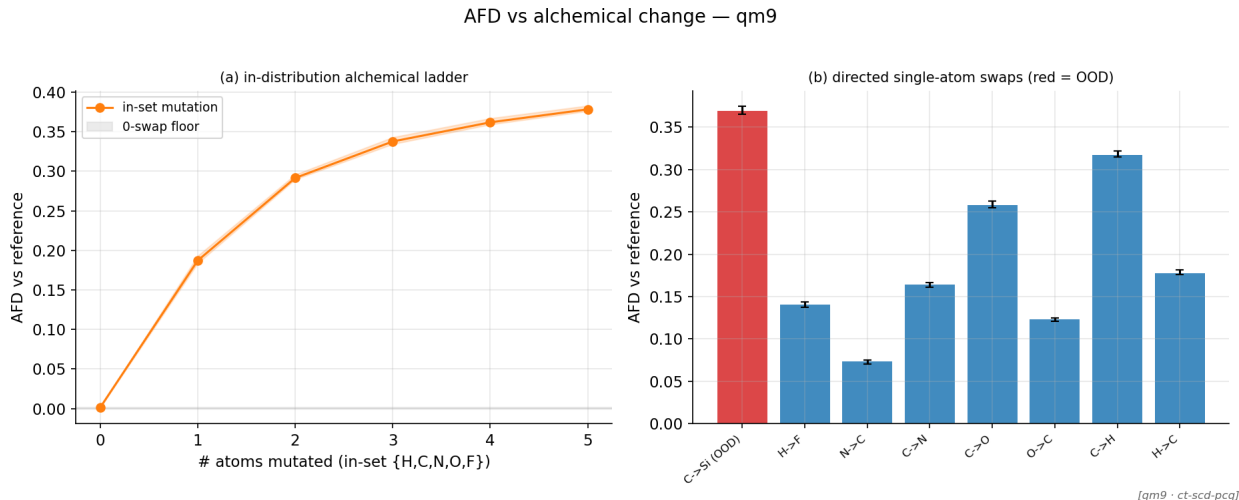


Figure 5: QM9 alchemical change: (a) integer in-set mutation ladder; (b) directed single-atom swaps along fixed element directions. The OOD C→Si swap (red) is the largest response.

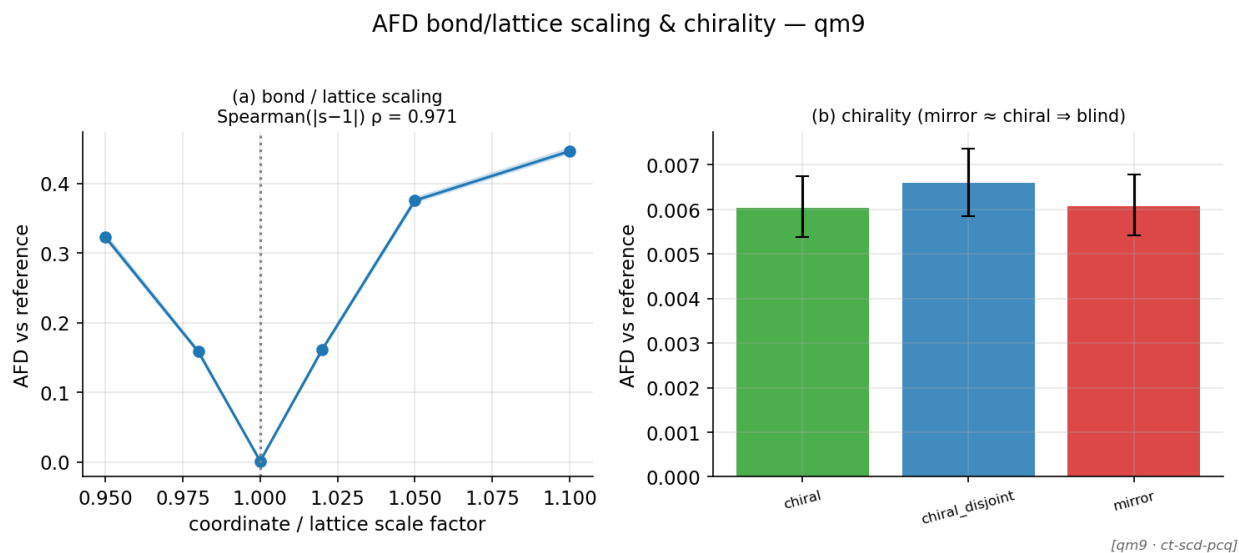


Figure 6: QM9 bonus tests: (a) AFD vs. uniform coordinate scale (V-shaped about 1.0); (b) chirality — a chiral fold, a disjoint chiral fold, and the mirror image of the first all score alike, confirming AFD is reflection-invariant.

2.3 Materials sensitivity (mp20)

Periodic crystals behave the same way once positions are wrapped into the (non-orthographic) unit cell (Figure 7): AFD is strongly sensitive and monotone ($\rho = 1.00$), detecting $\sigma \approx 0.01$ Å (0.022) and rising to 0.77 at $\sigma = 0.1$ Å; one mutated atom scores 0.22 and one vacancy 0.17, with transuranic OOD substitutions again above in-set (0.85 vs. 0.63 at 5 atoms). Isotropic lattice strain is registered as expected (Figure 8a), but **unit-cell repetition** exposes an invariance (Figure 8b): tiling a crystal into an $8\times$ supercell leaves AFD near the floor ($0.0049 \rightarrow 0.0070$), because the pooled embedding is effectively intensive. AFD correctly ignores supercell convention but cannot

flag a wrong cell multiplicity.

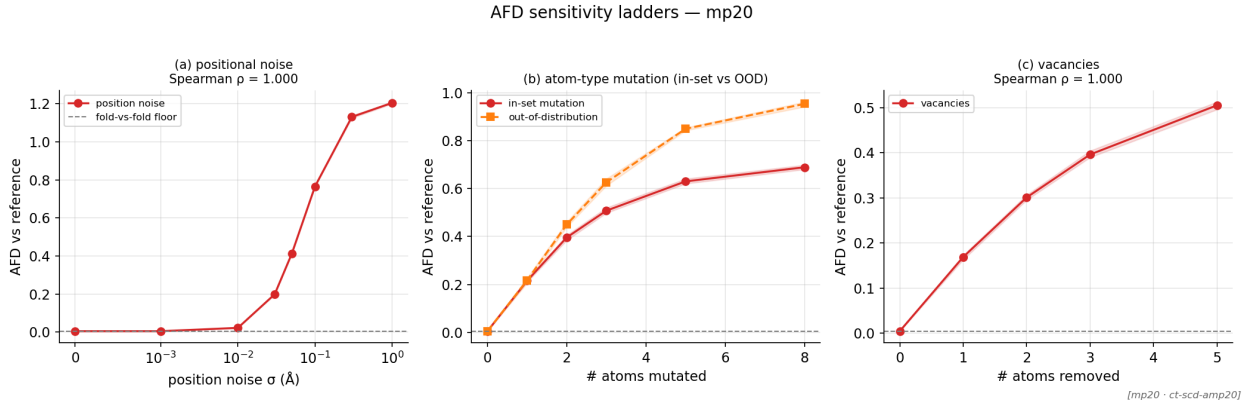


Figure 7: mp20 sensitivity ladders: wrapped position noise, atom mutation (in-set vs. transuranic OOD), and vacancies vs. the fold floor. Comparable sensitivity to QM9 under the matched checkpoint.

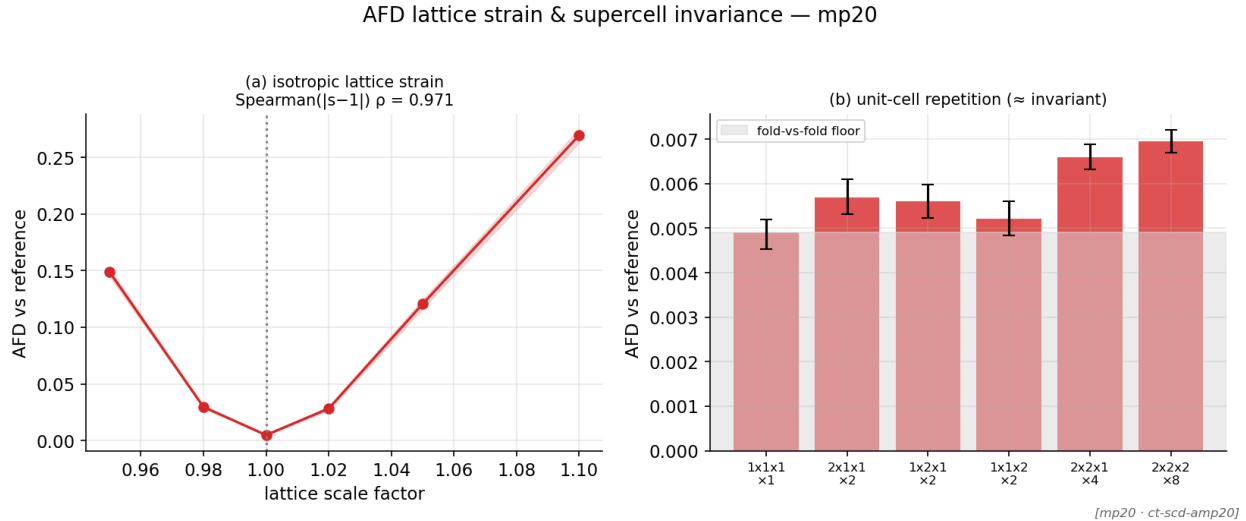


Figure 8: mp20 (a) isotropic lattice strain (V-shaped about 1.0) and (b) unit-cell repetition $\times 1$ – $\times 8$: AFD stays near the floor under supercell tiling — an invariance of the intensive embedding.

2.4 Calibration: reading a raw AFD value

Inverting the monotone QM9 ladders converts a raw AFD number into an equivalent physical degradation of ground truth (Table 1). As a rule of thumb, an **AFD of 0.1 corresponds to about 0.023 Å of coordinate jitter, or roughly half an atom per molecule mutated or missing**; a single mutated atom registers ≈ 0.19 (in-set) / ≈ 0.29 (OOD), a single vacancy ≈ 0.20 .

Table 1: AFD value $\rightarrow$ equivalent QM9 perturbation, by interpolation on the sensitivity ladders. A ground-truth interpretation aid, not generator scores.

AFD level	jitter σ (Å)	# atoms mutated	# atoms removed
0.01	0.011	0.05	0.05
0.05	0.016	0.26	0.25
0.10	0.023	0.53	0.51
0.20	0.036	1.12	1.04
0.30	0.047	2.18	2.15

2.5 Checkpoint dependence

Because the feature extractor is a pretrained CT, the checkpoint is part of the metric. Figures 9 and 10 rerun the position and mutation ladders across checkpoints. The in-domain specialist dominates every cross-domain alternative: at $\sigma \approx 0.1$ Å, QM9 scores 0.44 under `pcq` but only 0.15 under the materials checkpoint, and `mp20` scores 0.76 under `amp20` but 0.004 under `pcq`. Across all four checkpoints (Figure 10) the ordering is unambiguous — the matched specialist $\gg$ the generalist `all` $>$ `omo125` — so AFD must always be run with the domain-matched checkpoint.

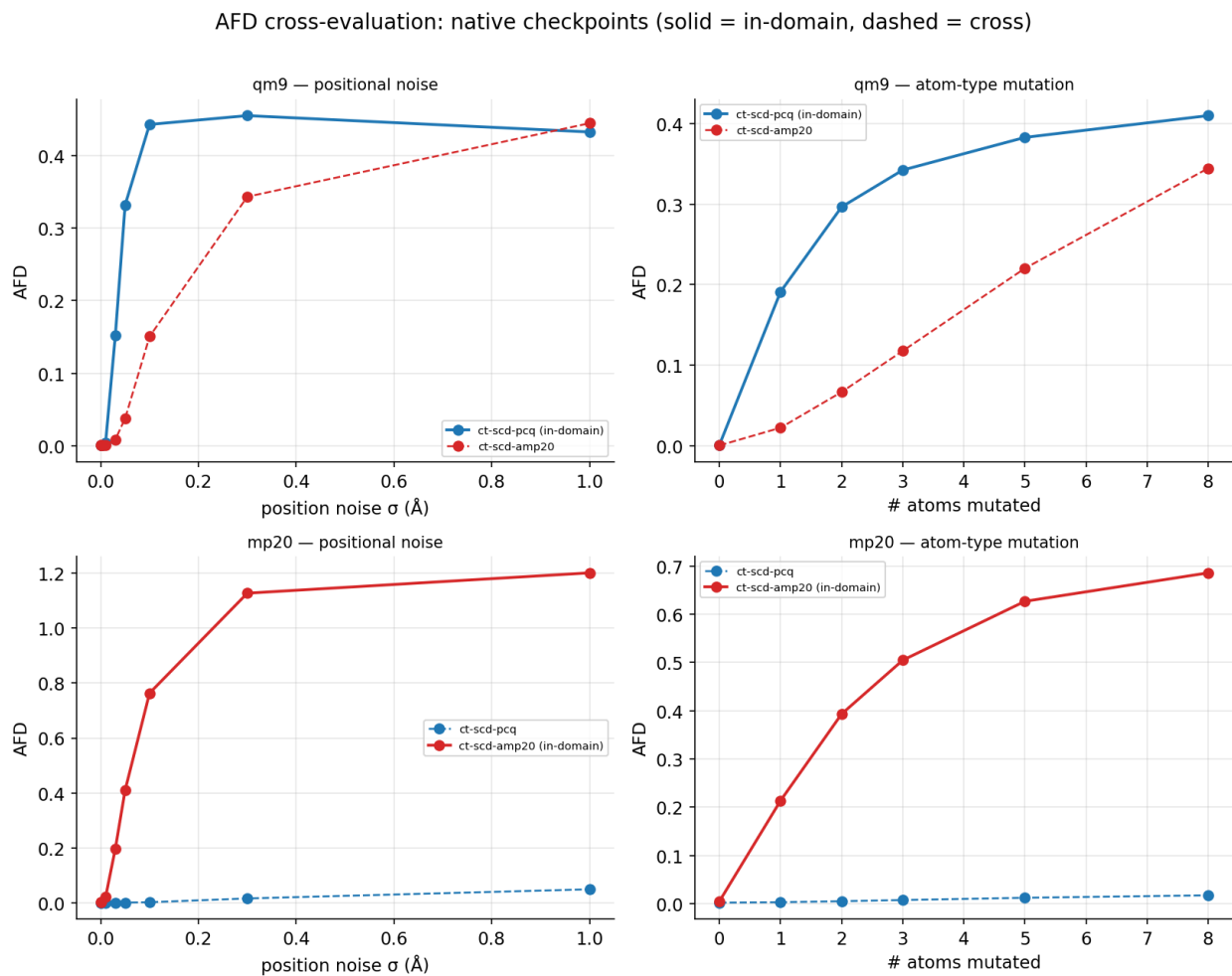


Figure 9: In-domain checkpoint (solid) vs. the other native checkpoint (dashed) for each dataset $\times$ {position, mutation}. The in-domain curve dominates everywhere.

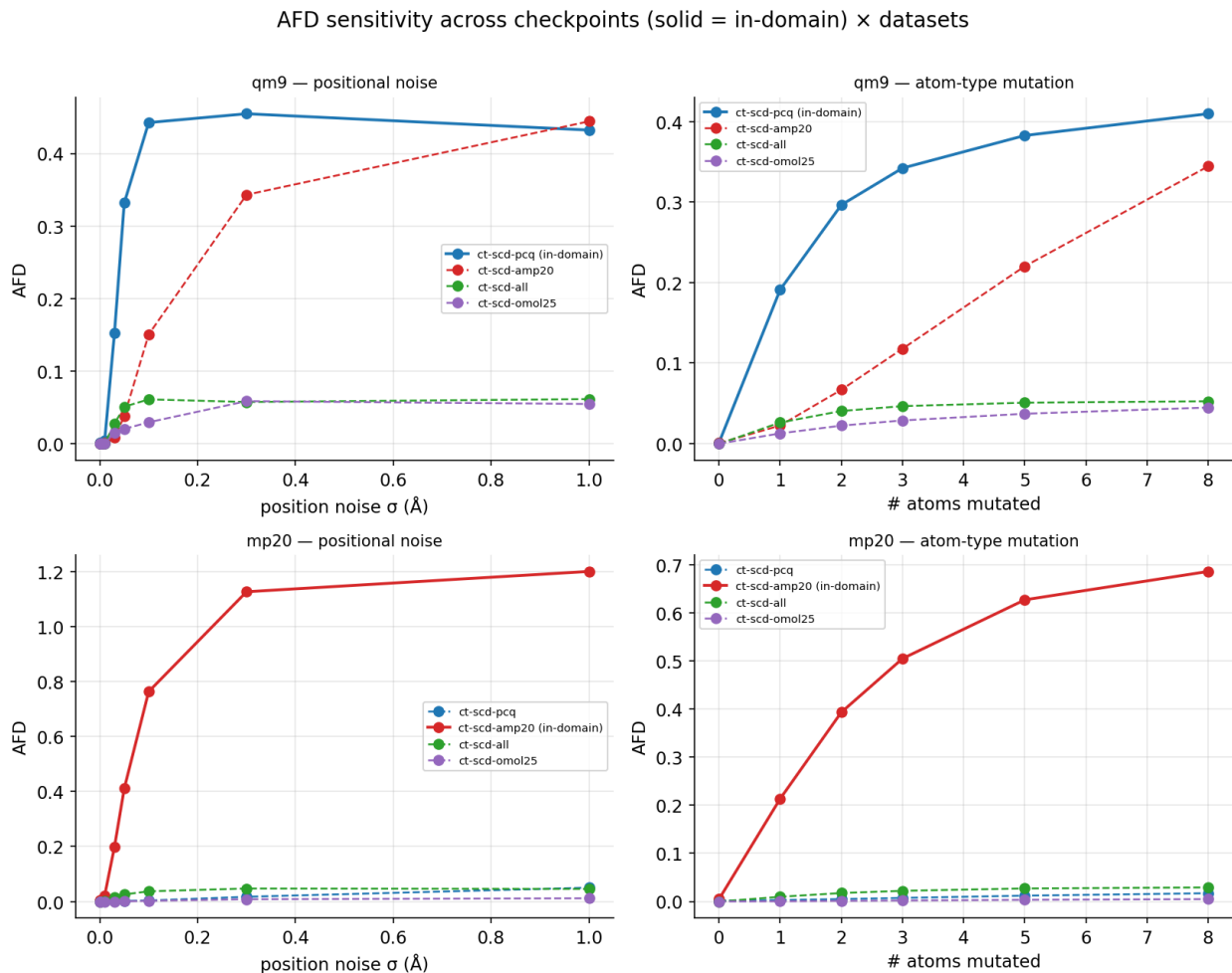


Figure 10: All four checkpoints on the same 2×2 grid. The domain-matched specialist is far more sensitive; the generalist `all` is a weak runner-up and `omol25` is weak on both modalities.

2.6 Application: scoring relaxed geometries

As a downstream check, AFD reads systematic geometry error and ranks post-processing steps (Figure 11). RDKit force-field conformers of *chemically correct* molecules already sit far above the floor (≈ 0.19) purely from geometry. Relaxing them orders the relaxers correctly: GFN2-xTB pulls the distribution *toward* the DFT reference (≈ 0.106), while the cheaper GFN-FF pushes it *further away* (≈ 0.242). AFD registers both the improvement and the regression.

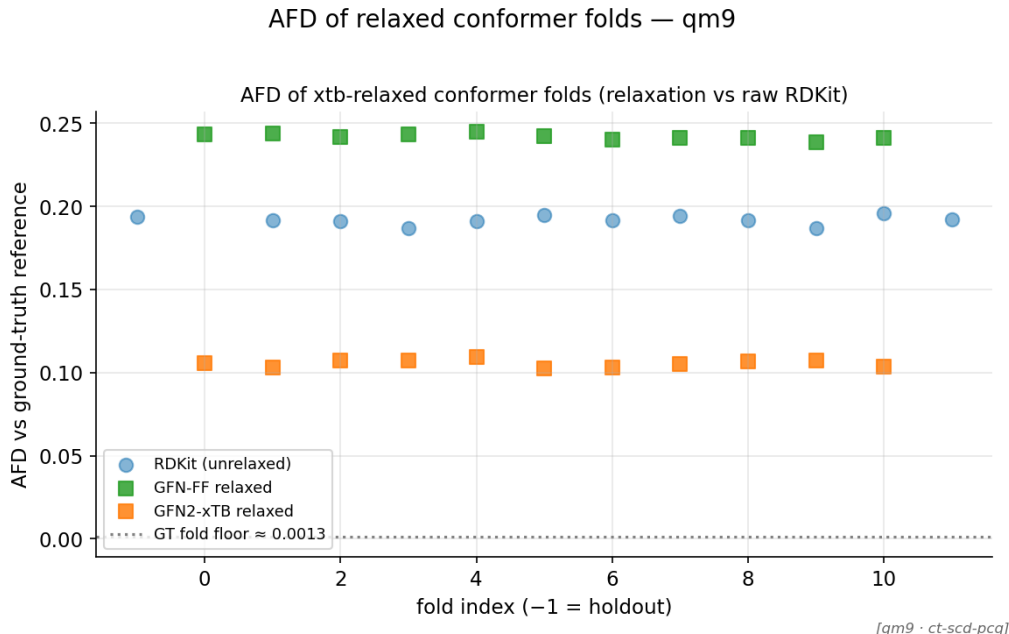


Figure 11: AFD of RDKit conformers relaxed with GFN-FF and GFN2-xTB vs. the DFT-geometry QM9 reference, against raw RDKit and the floor. GFN2-xTB improves on RDKit; GFN-FF regresses.

3 AFD Results

3.1 AFD scores for existing generative models

Figures 12–14 plot AFD against average sampling cost for a panel of published generators on each dataset (log–log; lower AFD is better, and points toward the lower-left are both faster and more faithful). Models with no timing metadata are drawn as labeled dashed reference lines. On QM9 (Figure 12) the strongest distributional fidelity comes from GCDM and the larger Zatom-J variants (AFD $\approx$ 0.015–0.05), while ADiT and EDM trail near 0.22–0.36; several fast samplers (e.g. the **tabasco** and small Zatom models) occupy the low-cost region at moderate AFD. On mp20 (Figure 13) the MatterGen variants achieve the lowest AFD (0.023–0.05) at high per-sample cost, whereas the invalid-cell-prone **Compositional** baseline sits worst ($\approx$ 1.19); the Zatom and DiffCSP families span the middle. On GEOM-drugs (Figure 14) **zatom80M-Jgeom** gives the best AFD (0.22) at by far the lowest cost, with EQGAT next and GeoLDM/ADiT-author highest. The exact values and timings are collected in Tables 2–4.

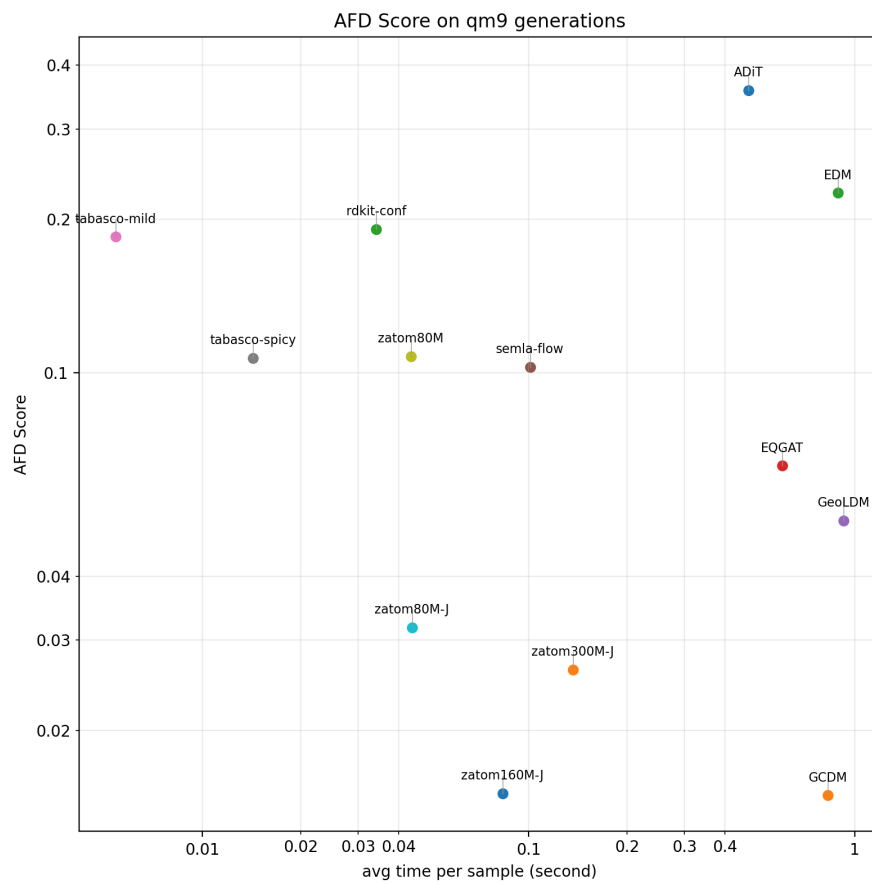


Figure 12: AFD vs. average time per sample on QM9 generations.

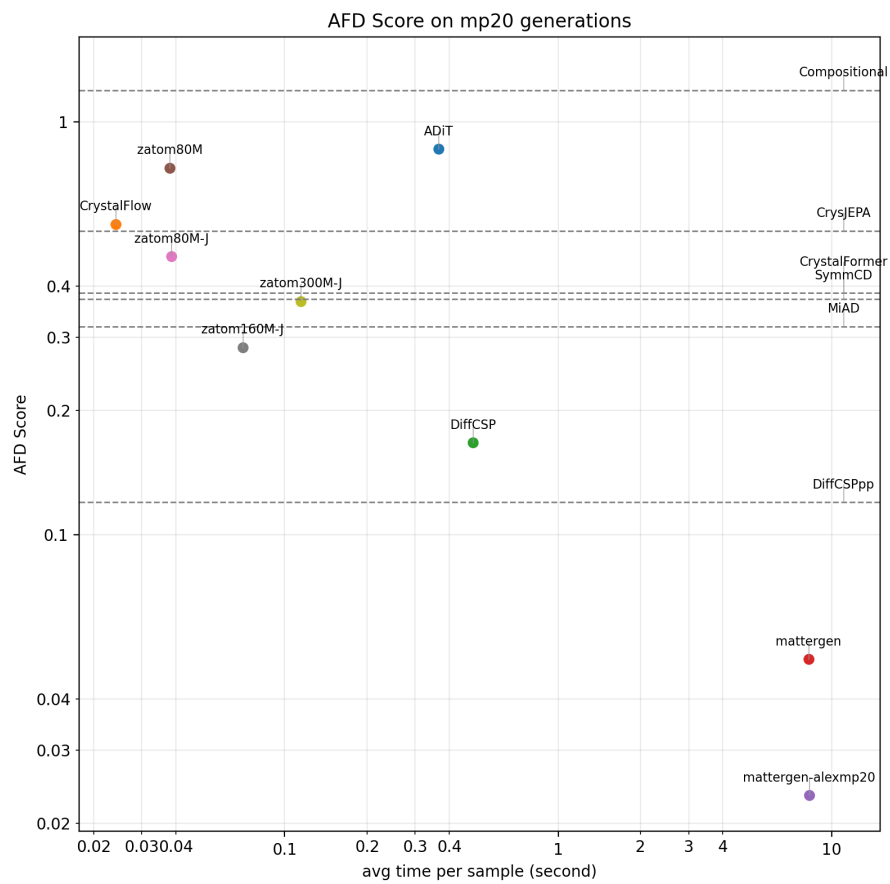


Figure 13: AFD vs. average time per sample on mp20 generations (the mp20 and mp20_alt model sets merged).

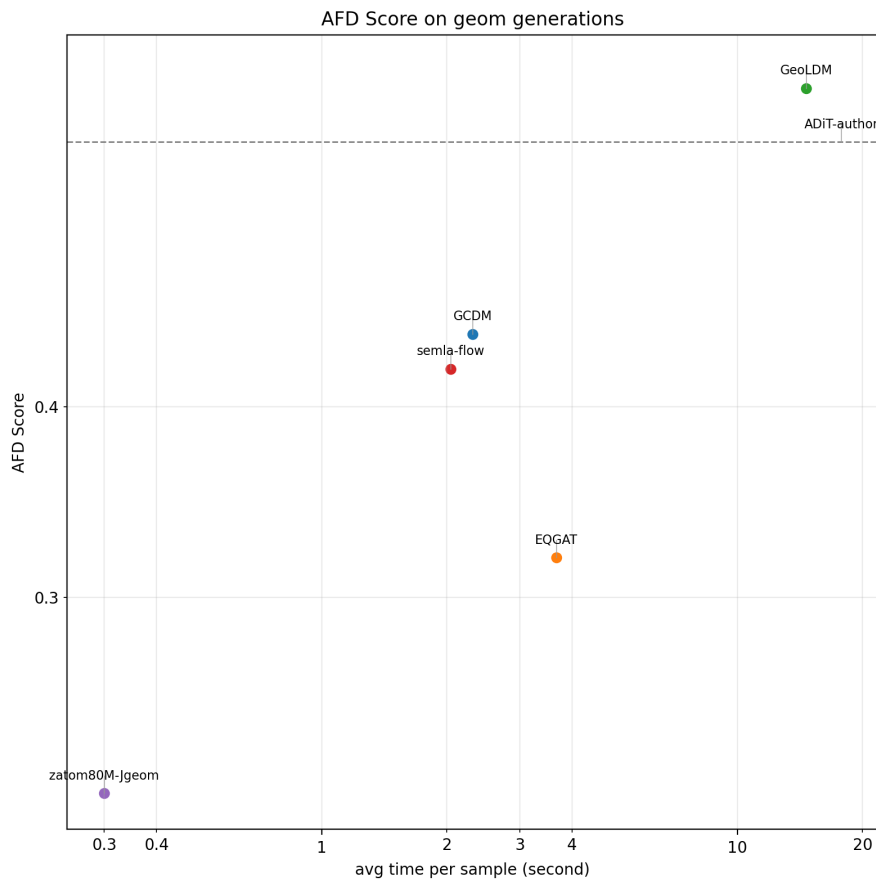


Figure 14: AFD vs. average time per sample on GEOM-drugs generations.

Tables 2–4 report, per generation set, the AFD score and the average time per sample. Lower is better in every column; the **best** value per column is bold and the second best underlined, and rows are ordered by AFD score from highest (worst) at the top to lowest (best) at the bottom.

Table 2: QM9 generation scores. Lower is better; **best** / second best per column.

Model	AFD score	time/sample (s)
ADiT	0.3569	0.4709
EDM	0.2249	0.8875
rdkit-conf	0.1911	0.03408
tabasco-mild	0.1845	0.005421
zatom80M	0.1078	0.04352
tabasco-spicy	0.1070	<u>0.01428</u>
semia-flow	0.1027	0.101
EQGAT	0.0659	0.5992
GeoLDM	0.0514	0.9238
zatom80M-J	0.0318	0.04389
zatom300M-J	0.0263	0.1365
zatom160M-J	<u>0.0150</u>	0.08316
GCDM	0.0149	0.8257

Table 3: mp20 generation scores (mp20 and mp20_alt merged). Lower is better; **best** / second best per column.

Model	AFD score	time/sample (s)
Compositional	1.1899	—
ADiT	0.8578	0.3655
zatom80M	0.7732	<u>0.03798</u>
CrystalFlow	0.5643	0.02408
CrysJEPA	0.5440	—
zatom80M-J	0.4714	0.0385
CrystalFormer	0.3852	—
SymmCD	0.3722	—
zatom300M-J	0.3670	0.1147
MiAD	0.3189	—
zatom160M-J	0.2838	0.07021
DiffCSP	0.1669	0.4885
DiffCSPpp	0.1197	—
mattergen	<u>0.0499</u>	8.227
mattergen-alexmp20	0.0234	8.267

Table 4: GEOM-drugs generation scores. Lower is better; **best** / second best per column.

Model	AFD score	time/sample (s)
GeoLDM	0.6456	14.63
ADiT-author	0.5955	—
GCDM	0.4460	2.304
semLa-flow	0.4232	<u>2.044</u>
EQGAT	<u>0.3187</u>	3.669
zatom80M-Jgeom	0.2235	0.2994

3.2 Legacy evaluations

The tables below report a separate, established evaluation suite (validity, uniqueness, novelty, Fréchet ChemNet Distance for molecules, and MLIP-based stability/energy/SUN for materials). **These results are preliminary and subject to change.** Rows are ordered worst (top) to best (bottom) by a composite mean-rank across the direction-corrected metrics; **best** / second best per column are highlighted among generated models, and ground-truth (GT) rows are shown for reference but excluded from highlighting. Per-molecule PoseBusters geometry checks are near-identical to the **valid** fraction and are omitted for brevity.

On QM9 (Table 5) the RDKit-conformer control is essentially GT-level, and among true generators the Zatom-J models and EDM lead the composite; EDM pays for its high novelty with the lowest validity, illustrating the validity/novelty trade-off.

Table 5: Legacy QM9 molecule metrics (full $\sim 10k$ sets, GT matched at 10k). FCD lower is better; others higher is better.

Model	rank	valid	novel	unique	FCD
EDM	11.73	0.766	0.606	<u>0.990</u>	0.648
tabasco-spicy	11.73	0.829	0.190	0.958	0.432
tabasco-mild	11.45	0.814	0.281	0.973	0.372
sempla-flow	9.91	0.899	0.187	0.840	1.085
zatom80M	9.64	0.860	0.192	0.971	<u>0.098</u>
GCDM	8.82	0.871	<u>0.463</u>	0.981	0.426
GeoLDM	7.36	0.907	0.420	0.979	0.192
EQGAT	6.64	0.910	0.410	0.977	0.164
zatom300M-J	6.64	0.919	0.135	0.964	0.122
ADiT	6.09	0.921	0.239	0.966	0.177
zatom80M-J	5.36	0.922	0.150	0.963	0.112
zatom160M-J	4.45	<u>0.936</u>	0.109	0.968	0.105
rdkit-conf	2.27	0.954	0.001	1.000	0.048
qm9_GT (GT)	1.90	0.946	0.002	1.000	0.050

On mp20 (Table 6) the picture is more mixed: MatterGen and MiAD lead the composite on validity and hull energy, while the Zatom-J models top the (meta)stability and SUN/MSUN columns; the **Compositional** baseline is near-perfect on composition validity but fails structural validity and hull energy.

Table 6: Legacy mp20 materials metrics (full $\sim 10k$ lightweight; MLIP stability/SUN on 2,000 samples). **e_hull** lower is better; others higher is better.

Model	rank	valid	unique	nov	stable	metastab.	e_hull	sun	msun
SymmCD	12.27	0.735	0.987	0.964	0.021	0.116	3.580	0.019	0.081
Compositional	11.91	0.220	0.953	1.000	0.004	0.004	12.522	0.004	0.001
CrystalFormer	11.27	0.699	0.996	0.981	0.021	0.160	7.697	0.020	0.117
CrystalFlow	10.09	0.872	0.990	1.000	0.018	0.038	1.141	0.018	0.019
CrysJEPA	9.82	0.964	0.449	<u>1.000</u>	0.018	0.041	0.649	0.011	0.015
zatom80M	9.82	0.725	0.933	0.998	0.018	0.161	2.681	0.018	0.139
DiffCSPpp	9.18	0.948	0.994	0.987	0.024	0.144	0.610	0.024	0.111
ADiT	8.82	0.906	0.963	0.999	0.028	0.066	0.853	0.028	0.035
zatom80M-J	8.00	0.885	0.901	0.993	<u>0.034</u>	0.280	0.891	<u>0.033</u>	0.243
mattergen-alexmp20	7.91	0.943	1.000	0.997	0.013	0.347	0.141	0.012	0.327
DiffCSP	6.64	0.952	0.998	1.000	0.021	0.153	0.383	0.021	0.132
zatom300M-J	6.36	0.943	0.879	0.990	0.038	<u>0.374</u>	0.421	0.037	<u>0.328</u>
zatom160M-J	6.32	0.952	0.874	0.991	0.032	0.417	0.408	0.031	0.379
MiAD	5.77	0.959	0.966	0.998	0.031	0.304	0.424	0.031	0.272
mattergen	5.64	<u>0.961</u>	<u>0.999</u>	0.998	0.029	0.197	<u>0.250</u>	0.029	0.167
mp20_GT (GT)	1.57	0.984	1.000	0.000	0.081	0.627	0.088	0.016	0.043

For GEOM-drugs (Table 7) we report the OpenBabel construction backend, which builds $\sim 98-99\%$ of point clouds and matches published validity numbers (GT ≈ 0.997); the alternative RDKit **DetermineBonds** backend under-reports GEOM validity because bond perception fails on many large drug molecules (a construction limitation, not a geometry defect), so those numbers are not reported here. Under OpenBabel, ADiT-author gives the best FCD and GeoLDM the highest raw

validity.

Table 7: Legacy GEOM-drugs molecule metrics (OpenBabel construction backend, $N = 3,000$). FCD lower is better; others higher is better.

Model	rank	valid	novel	unique	FCD
EQGAT-err	6.82	0.915	0.999	<u>1.000</u>	5.990
semLa-flow	6.82	0.922	0.989	0.991	8.260
GCDM	4.82	0.969	1.000	1.000	13.490
zatom80M-Jgeom	4.73	0.974	0.985	1.000	<u>3.781</u>
GeoLDM	4.36	0.982	<u>0.999</u>	1.000	21.529
ADiT-author	3.73	<u>0.981</u>	0.986	1.000	2.941
geom_GT (GT)	1.50	0.998	0.078	1.000	0.916